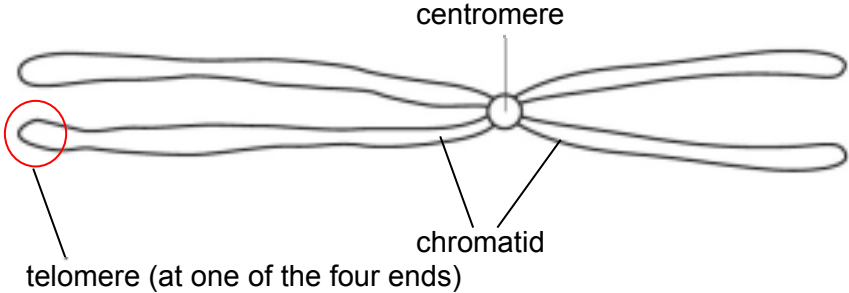


2(a)	P – chromosomes, arranged randomly / AW <i>or</i> <i>idea that</i> chromosomes present but not (yet) visible as consisting of two sister chromatids (joined by centromere) ; A description related to idea of chromosomes becoming visible / chromatin condensing Q – chromosomes are, arranged / aligned / AW, at the, (cell / spindle) equator / metaphase plate ; A equatorial plate R – no chromosomes visible / nucleolus is present / chromatin not condensed / euchromatin present ;	3
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Question	Answer	Marks
2(b)(i)	<i>drawing</i> two chromatids of identical length joined at the centromere ; <i>labels</i> chromatid ; <i>one label line is sufficient</i> R if second line drawn to arm of sister chromatid centromere ; telomere ; 	4
2(b)(ii)	histone(s) ; A highly basic (protein)	1
2(b)(iii)	<i>accept spindle fibres / microtubules as alternative to spindle</i> <i>max 2 if stages are named and are incorrect</i> <i>any three from:</i> (prophase) attachment to centromere of each chromosome ; A ref. to kinetochore in correct context (prophase) arranging / orientating / AW, chromosomes at the, equator / metaphase plate / AW ; A ref. to metaphase in correct context (anaphase) centromere division / separation of sister chromatids ; (anaphase) pulling, chromatids / daughter chromosomes, apart / towards <u>poles</u> ; (because) spindle fibres contract / (as) lengths of microtubules decreases / microtubules are disassembled ; (elongates in telophase to) push, poles / nuclei, apart ;	3